

BASIC ELECTRICAL MATERIALS AND METHODS

The exterior electrical system was privatized on 1 Mar 2013. With few exceptions the high-voltage electrical system is the property of the privatization contractor. Generally the point of demarcation is the low-voltage lugs of the supply transformer. Any exterior work to be done must be coordinated through the Base CE representative.

Voltage: Primary voltage is 13.2 kV and most secondary voltage systems are 120/208V or 277/480V, 3 phase, 4 wire. System voltage in each building must be verified.

Material and Equipment: Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years.

Regulatory Requirements: Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70. Electrical installations must conform to IEEE C2, NFPA 70, and requirements specified herein.

Warning Signs: Provide warning signs for the enclosures of electrical equipment including substations, pad-mounted transformers, pad-mounted switches, generators, and switchgear having a nominal rating exceeding 600 volts.

Motors – Single Phase: Unless otherwise requested, single-phase fractional-horsepower alternating-current motors must be high efficiency types.

Motors – Polyphase: Unless otherwise requested, polyphase motors must be selected based on high efficiency characteristics relative to the applications.

Painting - Factory Applied: Electrical equipment must only have factory-applied paint.

THREE-PHASE PAD-MOUNTED TRANSFORMERSCompartments - High Voltage, Dead-Front:

High-voltage compartment must contain the incoming line, insulated high-voltage load-break connectors, bushing well inserts, six high-voltage bushing wells configured for loop feed application, load-break switch handle(s), access to oil-immersed fuses, dead-front surge arresters, tap changer handle, connector parking stands with insulated standoff bushings, and ground pad.

- a. Insulated high-voltage load-break connectors: rated 95 kV BIL. Current rating: 200 amperes rms continuous. Connector must have a steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material.
- b. Bushing well inserts and feed-thru inserts: 200 amperes, 15 kV class. Provide a bushing well insert for each bushing well unless indicated otherwise.

Transformer:

Oil-insulated, two winding, 60 hertz, 65 ° C rise above a 30 ° C average ambient, self-cooled type. Copper or aluminum windings are acceptable.

Insulating Liquid:

Mineral oil: Type II, Provide identification of transformer as "non-PCB" and "Type II mineral oil" on the nameplate.

Corrosion Protection:

Bases and cabinets of transformers must be corrosion resistant and must be fabricated of stainless steel. Base must include any part of pad-mounted transformer that is within 3 inches of concrete pad. Paint bases, cabinets, and tanks Base Standard color.

Ground Ring for Equipment:

Provide a ground ring around high-voltage equipment consisting of a ring with copper wire around the periphery with ground rods in each corner, a ground rod in the transformer window and two lead-ins from the ground ring. Connections to the ring must be made by exothermic welding or approved irreversible compression fittings

UNDERGROUND TRANSMISSION AND DISTRIBUTION

Rigid Metal Conduit: Galvanized steel, threaded type.

Rigid Metal Conduit, PVC Coated: Galvanized steel, threaded type, coated with a polyvinyl chloride (PVC) sheath bonded to the galvanized exterior surface, nominal 40 mils thick.

Plastic Conduit for Direct Burial: NEMA TC 2, EPC-40-PVC.

600 Volt Wires and Cables: Service entrance and direct buried conductors must be Type USE. Conductors in conduit other than service entrance must be Type THHN/THWN-2. Conductors No.10 AWG must be stranded. Conductors No.12 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded copper. All conductors must be copper.

Medium Voltage Cable: Cable for underground distribution system must be ozone resistant ethylene-propylene-rubber-insulated (EPR) cable or cross-linked-thermosetting-insulated (XLP) cable. Cable must have a polyvinyl chloride jacket. Single phase circuits must use 100% neutral. Three phase circuits may use 1/3 neutral.

Medium Voltage Cable Terminations: Provide terminations including stress control terminator, ground clamp, connectors, and lugs. Terminator must be the product of one manufacturer, suitable for the type and materials of the cable terminated. Furnish components in the form of a "UL listed" kit, including complete instructions which must be followed for assembly and installation.

Indoor Terminations: Indoor terminator must be cold-shrink type or heat shrinkable type.

Buried Warning & Identification Tape: Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried cable and conduit. Tape must be detectable by an electronic detection instrument.

Medium Voltage Cable Joints: Provide power cable joints (splices) suitable for continuous immersion in water and direct burial. Make joints only in accessible locations in manholes or handholes by using materials and methods specified and as designated by the written instructions of the cable and joint kit manufacturer(s).

INTERIOR DISTRIBUTION SYSTEM**ELECTRICAL ROOMS & CLOSETS**

Locate service entrance equipment and other major electrical equipment in a dedicated electrical equipment room. Also provide electrical rooms and/or closets to separately house all other electrical equipment and components not normally found in occupied spaces including offices, conference rooms, laboratories, classrooms and auditoriums. REFER TO UFC 3-520-01.

MATERIALS AND EQUIPMENT

Materials, equipment, and devices must, as a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70. Grounding and bonding must be in accordance with NFPA 70 and NFPA 780.

CONDUIT AND FITTINGS: must conform to the following:

Rigid Metallic Conduit: GRS or rigid non-metallic must be used in concrete, masonry walls if grouted and slabs, exposed to the weather or where subject to physical damage and on exterior of buildings. PVC may also be used in concrete.

Rigid, Threaded Zinc-Coated Steel Conduit: NEMA C80.1

Rigid Nonmetallic Conduit: PVC Type EPC-40, or fiberglass conduit. Use for service entrance conduits from the utility substation or underground below floor slabs or when required for 400 Hz circuits. Do not use PVC above the first floor slab of buildings. Underground portions outside the building must be encased in minimum 3 inches of concrete and installed 36 inches below grade for high voltage conduits.

Intermediate Metal Conduit (IMC): IMC may be used in lieu of GRS as allowed by NFPA 70 and approved by Base Civil Engineering.

See UFC 3-520-01 for locations where MC cable is authorized.

Electrical, Zinc-Coated Steel Metallic Tubing (EMT): Use EMT for branch circuits and feeders above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar or grout, in hazardous locations, outdoors or in fire pump rooms.

Plastic-Coated Rigid Steel and IMC Conduit: Use in corrosive environments.

Flexible Metal Conduit: UL 1.

Liquid-Tight Flexible Metal Conduit, Steel: UL 360.

Fittings for Metal Conduit, EMT, and Flexible Metal Conduit: Ferrous fittings must be cadmium or zinc-coated.

Fittings for Rigid Metal Conduit and IMC: Threaded-type. Split couplings unacceptable.

Fittings for EMT: Connections must be threadless, die cast and raintight. Do not use crimp-on type. Set screw type may be used indoors only.

Fittings for Rigid Nonmetallic Conduit: NEMA TC 3, UL 514B.

Surface Metal Raceway:

Two-piece painted steel, totally enclosed, snap-cover type.

Cable Trays:

Cable trays must form a wireway system and must be of depth needed. Cable trays must be constructed of steel that has been zinc-coated after fabrication.

Outlet Boxes & Covers:

UL 514A, cadmium- or zinc-coated, if ferrous metal, or UL 514C, if nonmetallic.

Cabinets, Junction Boxes, & Pull Boxes:

Volume greater than 100 cubic inches, hot-dip, zinc-coated, if sheet steel.

Wires and Cables: Wires and cables must meet applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Wires and cables manufactured more than 12 months prior to date of delivery to site must not be used.

Conductors: Conductors No. 8 AWG and larger diameter must be stranded. Conductors No. 12 AWG and smaller diameter must be solid, except that conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3, must be stranded unless specifically indicated otherwise. Conductors No. 10 must be stranded. Conductor sizes and ampacities shown are based on copper, unless indicated otherwise. All conductors must be copper.

Minimum Conductor Sizes: Minimum size for branch circuits must be No. 12 AWG; for Class 1 remote-control and signal circuits, No. 14 AWG; for Class 2 low-energy, remote-control and signal circuits, No. 16 AWG; and for Class 3 low-energy, remote-control, alarm and signal circuits, No. 22 AWG.

Wire Insulation: Unless specified or indicated otherwise or required by NFPA 70, power and lighting wires must be 600-volt, Type THHN/THWN-2. Grounding wire may be type THHN/THWN-2. Remote-control and signal circuits must be Type THHN/THWN.

Toggle Switches: Single Pole, double pole, three-way, and four-way, totally enclosed with bodies of thermoplastic and/or thermoset plastic and mounting strap with grounding screw. Wiring terminals must be screw-type, side-wired. Voltage rating must be 120V/277V and 20 amp. Switches must be specification grade as a minimum.

Receptacles: Hard use, heavy-duty, grounding-type. Face and body must be thermoplastic supported on a metal mounting strap. Provide screw-type, side-wired wiring terminals. Connect grounding pole to mounting strap. Receptacles must be specification grade as a minimum.

Weatherproof Receptacles: Provide in cast metal box with gasketed, weatherproof, cast-metal cover plate and gasketed cap over each receptacle opening. Receptacle must be UL listed for use in "wet locations with plug in use."

GFCI Receptacles: Ground-Fault Circuit Interrupter Receptacles must be duplex type, 20 A, for mounting in standard outlet box.

Range Receptacles: NEMA 14-50 configuration, rated 50 amperes, 125/250 volts.

Dryer Receptacles: NEMA 14-30 configuration, rated 30 amperes, 125/250 volts.

Panelboards: Must have a short-circuit current rating of 10,000 amperes symmetrical minimum or higher rating as needed for short-circuit current at the point of application. Panelboards for use as service disconnecting means must be listed for such use. Panelboards must be circuit breaker-equipped. Design must be such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL. Directories must indicate load served by each circuit in panelboard. Type directories and mount in holder behind transparent protective covering. Load centers must not be used. Circuit breakers must be bolt-on type. Do not use dual section panelboards. Do not use series rated breakers. Busses in panelboards must be copper.

Panelboard Neutrals for Non-Linear Loads: Panelboard type must have been specifically UL heat rise tested for use on non-linear loads. Panelboard must be heat rise tested except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing. Two neutral assemblies paralleled together with cable are not acceptable. Nameplates for panelboard rated for use on non-linear loads must be marked "SUITABLE FOR NON-LINEAR LOADS."

Circuit Breakers: Thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker must be mounted or must be rated for the short-circuit availability at the point of connection of the panelboard. Series rated circuit breakers and plug-in circuit breakers are not acceptable. Electronic type circuit breakers are acceptable.

Circuit Breaker With GFCI or GFI: Must be UL 943 and NFPA 70 listed.

Arc-Fault Circuit-Interrupters: Molded case circuit breaker must be rated as indicated.

400 Hz Panelboard & Breakers: Panelboards and breakers in 400 Hz systems must be "400 Hz" rated and labeled.

Load Centers: Load centers must not be used.

Transformers: General purpose, dry-type, self-cooled. Provide transformers in NEMA enclosure. Transformer must have 220 ° C insulation system for transformers 15 kVA and greater, and must have 180 ° C insulation for transformers rated 10 kVA and less, with temperature rise not exceeding 80 ° C under full-rated load in maximum ambient of 40 ° C. Transformer of 80 ° C temperature rise must be capable of carrying continuously 130 percent of nameplate kVA without exceeding insulation rating. Transformers must be quiet type with maximum sound level at least 3 decibels less than NEMA standard level for transformer ratings indicated. K rated transformers must be used to mitigate harmonic effects of non-linear loads. Copper or aluminum windings are acceptable. Provide energy efficient transformer.

Motors: Motors for operation on 208-volt, 3-phase circuits must have terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits must have terminal voltage rating of 460 volts.

TELEPHONE & LAN SYSTEM: COMPLY WITH EIA/TIA-568-B, 569 AND 606. Reference UFC 3-580-0. Coordinate closely with communications.

Grounding and Bonding Equipment: Provide equipment grounding conductors. Ground rods must be copper-clad steel, with minimum diameter of 3/4 in and minimum length of 10 ft.

INTERIOR LIGHTING: Reference UFC 3-530-01

Fluorescent Lighting Fixtures: Fluorescent fixtures must have electronic ballasts.

Fluorescent Lamps: T-8 rapid start low mercury lamps must be rated 32 watts (maximum), 2800 initial lumens (minimum), CRI of 75 (minimum), color temperature of 3500 K, and an average rated life of 20,000 hours. Low mercury lamps must have passed the EPA Toxicity Characteristic Leachate Procedure (TCLP) for mercury by using the lamp sample preparation procedure described in NEMA LL 1.

Provide program start ballast for those fixtures that are frequently switched. Normal fluorescent ballasts may be rapid start or instant start.

Compact Fluorescent Lamps: Compact fluorescent lamps must be: CRI 80, minimum, 3500 K, 10,000 hours average rated life, and as follows:

- a. T-4, twin tube, rated 5 watt, 250 initial lumens (minimum); 7 watts, 400 initial lumens (minimum); 9 watts, 600 initial lumens (minimum); and 13 watts, 825 initial lumens (minimum).
- b. T-4, double twin tube, rated 13 watts, 900 initial lumens (minimum); 18 watts, 1200 initial lumens (minimum); and 26 watts, 1800 initial lumens (minimum).

High-Intensity-Discharge (HID) Lighting Fixtures: Provide HID fixtures with tempered glass lenses when using metal-halide lamps. If approved, open Metal Halide may be used only if the lamp is rated for open fixture in industrial areas.

HID Ballasts: Must be constant wattage autotransformer (CWA) or regulator, high power factor type. Provide single-lamp ballasts which must have a minimum starting temperature of minus 30 degrees C.

HPS Ballasts: must have a solid-state igniter starter.

Standby High-Pressure Sodium (HPS) Lamps: Standby HPS lamps must have two arc tubes and an average rated life of 40,000 hours (minimum) and hot restart. Instant lumen output must be 8 percent, minimum, of total light output.

Metal-Halide Lamps: Use where color rendition is needed.

Incandescent Lighting Fixtures: UL 1598.

Incandescent Dimmer Switch: Single-pole, 120 volt ac, full-range rotary or slide, on-off type, with built-in electromagnetic interference filter.

Exit Signs: Exit signs must be self-powered type.

Self-Powered LED Type Exit Signs (Battery Backup):

Provide with automatic power failure device, test switch, pilot light, integral self-testing module and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed electrolyte type, must operate unattended, and require no maintenance, including no additional water, for a period of not less than 5 years. LED exit sign must have emergency run time of 1 1/2 hours (minimum) and be red.

Fluorescent Emergency Lighting System:

Each system must consist of an automatic power failure device, test switch operable from outside of the fixture, pilot light visible from outside the fixture, integral self-testing module and fully automatic solid-state charger in a self-contained power pack. Battery must operate unattended and require no maintenance, including no additional water, for a period of not less than 5 years. Emergency ballasts provided with fixtures containing solid-state ballasts must be fully compatible with the solid-state ballasts. Industrial areas may use wall-pack emergency lights.

Auxiliary Instant-On System:

UL listed, automatically switched instant-on lamp. Quartz lamp must come on when luminaire is initially energized and following a momentary power outage must remain on until HID lamp reaches approximately 60 percent light output.

Occupancy Sensors:

Provide combination ultrasonic-infrared occupancy sensors for classrooms and conference rooms and other locations where required for energy conservation.

LED Fixtures:

LED lighting, where appropriate, is permitted. Induction fluorescent lighting, where appropriate, is permitted.

Wireless Lighting Control Systems:

Wireless lighting control systems are not allowed to be used on any projects.

EXTERIOR LIGHTING Reference UFC 3-530-01

Poles: provide poles designed for wind loading of 90 miles per hour determined in accordance with AASHTO LTS-4 while supporting luminaires. REFERENCE UFC 3-301-01

Aluminum Poles: Provide aluminum poles manufactured of corrosion resistant aluminum alloys conforming to AASHTO LTS-4 for Alloy 6063-T6 or Alloy 6005-T5 for wrought alloys and Alloy 356-T4 (3, 5) for cast alloys. Poles must be seamless extruded square or round type with minimum wall thickness to meet loads.

Luminaires: Fixtures must be shoebox type compatible with existing fixtures on Base.

Lamps: Use HPS lamps except where color rendition is required. LED lighting, where appropriate, is permitted. Induction fluorescent lighting, where appropriate, is permitted.

HID Ballasts: Must be constant wattage autotransformer (CWA) or regulator, high power factor type. Provide single-lamp ballasts which have a minimum starting temperature of minus 30 degrees C. HPS ballasts must have a solid-state igniter starter.

Photocell Switches: provide hermetically sealed cadmium-sulfide or silicon diode type cell. Switch must turn on at or below 3 foot-candles and off at 2 to 10 foot-candles. A time delay must prevent accidental switching from transient light sources. Provide switch in areas including:

- a. Parking areas must be controlled by common photocell.
- b. Exterior building lights with same on/off schedule must be controlled by common photocell.
- c. In a cast weatherproof aluminum housing with adjustable window slide, rated 1800 VA, minimum.

Generators and Transfer Switches: Reference UFC 3-540-01

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GENERAL

- a. All mechanical equipment shall be produced by a manufacturer that has a minimum of 5 years of history manufacturing the particular or similar equipment. Said equipment shall be a product normally produced by the manufacturer on a regular basis. The manufacturer shall have the required in-house technical expertise and testing capabilities required to support the products operational and performance requirements.
- b. Major mechanical equipment shall be factory tested prior to shipping. Testing and compliance must be documented and submitted. Final acceptable for any project shall not be provided without this testing data.
- c. All required items to make the equipment operational and useable shall be provided by the equipment manufacturer.
- d. All major mechanical equipment (to include but not limited to fans, air handling units, chillers, direct expansion equip, boilers, all terminal boxes (VAVs, FTBs, etc), preheat and reheat coils, generators, pumps and adjustable frequency drives) as a minimum, shall be included into the existing Base EMCS. Head end graphics must be updated with every renovation.
- e. All major mechanical equipment must have acceptable warranties. Warranties shall be included with submittals for approval. Include a minimum, 1 year labor, 5 year parts for all major equipment. All warranties shall be provided from the manufacturer of the equipment where possible, and not offered through third party contractors or installation companies.
- f. All underground piping shall have a tracer wire.
- g. Underground water pipe shall be buried to a depth of four feet and no deeper.
- h. Directional boring is permissible however, it must be accompanied by signed and sealed (by a licensed professional engineer) drawings and calculations, as well as comply with all applicable standards and UFCs. Pit bores are prohibited.
- i. Backflow preventers shall be installed no more than five feet above the floor and twelve inches from the wall.
- j. All new HVAC automated controls shall be DDC type. All DDC controls shall be capable of being controlled and monitored through the existing base EMCS (Johnson Metasys). Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.
- k. Constant primary, variable secondary pumping systems are preferred. Variable primary only pumping systems will be considered on a case-by-case basis.
- l. Cooling load and psychometric calculations must be provided at the 35% submittal to ensure the proper sizing of new equipment. 35% documents will not be approved until this document is reviewed and approved. These calculations are required for all HVAC projects.
- m. Final zoning of the HVAC system is the responsibility of the contractor and provided with the 65% submittal. The contractor must receive written approval from the government on the final zoning plan. 65% documents will not be approved until this document is reviewed and approved.
- n. Provide Outside Air (OA) calculations IAW UFC and ASHRAE 62.1 as a part of the 95% submittal. These calculations are required for all HVAC projects.

MECHANICAL INSULATION

Piping Insulation

- a. Domestic hot and cold water installed above ground, shall be insulated with a minimum 1/2 inch mineral fiber.
- b. Heating hot water shall be insulated with a minimum 1 inch mineral fiber.
- c. Chilled water and dual service water piping shall be insulated with a minimum 3/4 inch foam glass, phenolic foam, or fiberglass.
- d. Horizontal internal storm drainage piping runs shall be insulated with 1 inch mineral fiber.
- e. Exterior chilled water and heating water piping outdoors and above grade shall be insulated with 1 inch foam glass, phenolic foam, or closed cell glass/polyisocyanurate with an aluminum jacket.
- f. Buried exterior chilled water, heating water and dual service piping shall be factory fabricated, pre-insulated piping suitable for exterior installation. Insulation shall be urethane with 1 inch minimum thickness. All ferrous metal piping shall have pre-insulated, coated elbows and joints due to a high water table in an effort to mitigate rust and deterioration.
- g. Refrigerant piping insulation shall be black, closed cell flexible elastomeric insulation, 1 inch thick.
- h. Insulation shall comply with the above requirements or energy efficiency standards, whichever one is more stringent. At a minimum, the insulation thickness, type, and application shall comply with the most current editions of ASHRAE 90.1, MICA National Commercial and Industrial Insulation Standards, and International Energy Conservation Code (IECC).

Ductwork Insulation

- a. Concealed supply, return and outside air ductwork shall be insulated with 2-inch thick, 1-1/2 pounds per cubic foot blanket flexible mineral fiber (must meet or exceed an R-8 rating IAW International Energy Conservation Code C403.11.1). Insulation shall be external to the ductwork. Internally lined duct and fiber duct board (when not used as insulation) are not acceptable.
- b. Supply, return and outside air ductwork exposed in mechanical rooms shall be insulated with 1-inch thick, 3 pounds per cubic foot rigid mineral fiber board insulation. External insulation only.
- c. Ductwork exposed outdoors shall be insulated with 1-inch thick flexible cellular insulation and wrapped with a minimum of 27 gauge aluminum jacket. (must meet or exceed an R-8 rating IAW International Energy Conservation Code C403.11.1)
- d. Mastic-plus-embedded-fabric systems are the preferred joining methods for all joints, seams, and connections. Do not use pressure sensitive tape unless previously approved by the Contracting Officer (CO) and Contracting Officer's Representative (COR).
- e. Adhesives and installation of all insulation shall be in accordance with manufacturer's recommendations.
- f. Insulation shall comply with the above requirements or energy efficiency standards, whichever one is more stringent. Insulation thickness shall meet the minimums listed above or the current edition of ASHRAE 90.1, or IECC, whichever is more stringent.
- g. All insulation shall comply with NFPA 90A standards.

CHILLED, CONDENSER, OR DUAL SERVICE WATER PIPING

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- a. Chilled water and dual service piping 2-1/2 inches and larger shall be Schedule 40 black steel with welded joints. Random Copolymerized Polypropylene (PP-R) and Random Copolymerized Polypropylene High-Crystallinity Resin (PP-RCT) piping typical to Aquatherm "Blue Pipe" is an acceptable alternative to black steel, for both interior and exterior applications. When used in applications for chilled, dual service, or heating hot water, all piping shall be insulated (refer to previous sections for more information).
 - b. Chilled water and dual service piping 2 inches and smaller shall be type "L" hard drawn copper or PP-R/RCT.
 - c. Condenser water piping shall be Schedule 40 black steel or PP-R/RCT. Condenser water piping is not required to be insulated, but is required to be provided with UV protection. A minimum of 2-coats of Tnemec paint is required for all exposed black steel condenser water piping. Provide UV protection for the PP-R/RCT in accordance with the manufacturer's recommendations.
 - d. End connections to valves and equipment shall be threaded on piping 2 inches and smaller. Pipe ends 2-1/2 inches and larger shall be flanged.
 - e. External buried chilled water and dual service piping shall be pre-insulated black steel double wall type or PP-R/RCT. If black steel is chosen, the carrier pipe shall be Schedule 40 black steel. Insulation shall be polyurethane with 1-inch minimum thickness. External jacket piping shall be PVC or high density polyurethane. Insulation shall comply with the above requirements or energy efficiency standards, whichever one is more stringent.
 - f. Gate valves 2-inch and smaller shall be bronze in accordance with MSS SP-80. Gate valves 2-1/2 inch and larger shall be cast iron in accordance with MSS SP-70.
 - g. Ball valves 2-inch and smaller shall be copper alloy in accordance with MSS SP-110. Flanged ball valves 2-1/2 inches and larger shall be bronze in accordance with MSS SP-72.
 - h. Butterfly valves shall be cast iron flanged, Type "I" tight shutoff type in accordance with MSS SP-67.
 - i. Flow measuring devices shall be Venturi type and capable of being read/monitored by the EMCS.
 - j. All steel/metal piping used on CAFB projects shall be made in America.

PUMPS

- a. Pump size and type shall be selected based on maximum efficiency at operating point. Pump shall be non-overloading over range of pump curve.
- b. Base-mount, end-suction pumps are preferred for chilled water, heating water and dual service duty.
- c. Use double suction pumps on condenser water systems where possible.
- d. Use variable frequency drives on pumps in secondary piping and variable primary systems.
- e. Pumps shall start/stop manually and through the base EMCS.
- f. Flexible connectors shall be used at all pumps and equipment.
- g. Inlet and outlet pressure gauges shall be provided on all pumps.
- h. Provide one additional unused, and capped, pete's plug on the pump inlet and outlet of each pump.

- i. Differential pressure shall be provided and monitored at each pump. Equipment must be capable of being read on EMCS.
- j. Electric motors shall be NEMA MG-1, premium efficiency.
- k. Insulate chilled water pumps with removable, fabricated rigid insulation unit.

PLUMBING SYSTEMS

- a. Sanitary waste, vent and storm piping inside the building shall be Schedule 40 PVC. Sanitary waste and storm piping below grade and outside the building shall be PVC (SDR-35).
- b. Domestic water piping inside the building shall be type "L" hard drawn copper in accordance with ASTM B 88. Domestic water piping below slab shall be type "K" soft drawn copper.

PLUMBING FIXTURES

Water Closets: Elongated, wall mounted flush valve type. Provide ASME A 112.6.1M carrier with feet.

Urinals: Vitreous china, wall mounted, siphon jet with ASME A 112.6.1M concealed chair carrier.

Lavatory (Wall Hung): White enameled cast iron as per ASME A 112.19.1M. Provide ASME A 112.18.1M self-closing metering type push button faucets and perforated grid strainers. Faucets shall have insulated traps on wheelchair lavatories. Preferred faucet manufacturer is Delta, model XXXXX.

Lavatory (Counter Mounted): Oval enameled cast iron self rimming type. Self-closing metering push button type, center set faucets. Preferred faucet manufacturer is Delta, model XXXXX.

Service Sink: White enameled cast iron, wall mounted and floor supported by wall outlet cast iron P-trap. Provide ASME 112.18.1M copper alloy back mounted combination faucets with vacuum breaker and 3/4 inch external hose threads.

Electric Water Cooler: Wall mounted recessed, high/low dual bubbler style, air cooled condensing unit with 4.75 gph minimum capacity. Bubblers shall be controlled by push bars on each side and in front.

Water Heaters - Natural Gas: Commercial grade and in accordance with ASME Boiler and Pressure Vessel Code Section IV and ANSI Z21.10.3. Tanks shall be rated for 150 psi working pressure, ASME stamped and glass or cement lined. Heater shall be insulated with fiberglass insulation. Standby heat loss shall conform to ASHRAE 90.1 high efficiency requirements. Provide ASME "T and P" relief valve.

Water Heaters - Electric: Commercial grade and in accordance with ASME Boiler and Pressure Vessel Code Section IV and ANSI Z21.10.3. Tanks shall be rated for 150 psi working pressure, ASME stamped and glass or cement lined. Heater shall be insulated with fiberglass insulation. Standby heat loss shall conform to ASHRAE 90.1 high efficiency requirements. Provide ASME "T and P" relief valve.

LOW PRESSURE WATER HEATING BOILERS

- a. Horizontal steel fire tube type, package unit mounted on a skid-type structural steel base.
- b. Forced draft condensing type boiler complying with ASHRAE 90.1 and MIL-B-17452.
- c. Burner shall include interrupted pilot type ignition system or other energy efficient type. Combustion control system shall be the positioning type. Burner controls and safety equipment shall conform to requirements of MIL-B-18796.

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- d. Boiler trim shall comply with ASME Boiler and Pressure Vessel Code Section IV and MIL-B18796. Provide an emergency disconnect switch on wall outside boiler room entrance.
 - e. Provide draft and venting in accordance with manufacturer's recommendations.
 - f. Preferred boiler manufacturer is Cleaver Brooks.
 - g. Boiler shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

CENTRAL REFRIGERATION EQUIPMENT

CHILLERS: When building/space loads are over 20 tons, chillers are to be incorporated into the design. Otherwise, packaged DX unitary outdoor/rooftop equipment shall be used.

- a. 20 Tons to 70 Tons:
Air cooled, self-contained chillers with hermetically sealed scroll type compressors. A minimum of two compressor circuits shall be provided. Chiller shall be in accordance with ARI-550. Refrigerant shall have an Ozone Depletion Factor (ODF) of 0.05 or less. No more R-22 equipment shall be installed. Preferred refrigerants are R-134a and R-407c. Preferred manufacturer is Daikin/McQuay.
- b. 70 Tons to 200 Tons:
Water cooled chiller with helical rotary (screw – type) compressors. A minimum of two compressor circuits must be provided. Chiller shall be in accordance with ARI-550. Compressors shall be capable of operating stably at 10 percent of full load. Refrigerant shall have an Ozone Depletion Factor (ODF) of 0.05 or less. Chiller shall have factory installed purge system. Preferred manufacturer is Daikin/McQuay
- c. 200 Tons and Larger:
Water cooled centrifugal chillers. Compressors shall be capable of operating stably for indefinite periods of time at any stage of capacity from 100 to 10 percent of load. Chillers shall be in accordance with ARI-550. Chiller shall have factory installed purge system. Preferred manufacturer is Daikin/McQuay
- d. Chillers shall include factory control panels and starters that are compatible with the existing Base EMCS.
- e. Chillers shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.
- f. All chillers shall meet or exceed the efficiency ratings within the current editions of ASHRAE 90.1 and the IECC.

COOLING TOWERS

- a. Open distribution type and factory assembled in accordance with FS-A-A-59223. Constructed of appropriate material for the service conditions with PVC fill. FRP construction is preferred.
- b. Fan motor shall be direct drive with a variable frequency drive. Motor, fan, and controls shall be located out of evaporation air stream.
- c. Condenser water circuit shall include a bypass valve to dump bypass water into the tower sump when required.
- d. Provide automatic feed chemical treatment.
- e. Provide factory ladders and handrails.

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- f. Make-up water shall have bubble type full shut-off valve.
 - g. Provide factory installed air inlet screens to prevent birds and trash from entering tower. Screen material shall be stainless steel.
 - h. Cooling towers shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

AIR HANDLING EQUIPMENT

AIR HANDLING UNITS

- a. Air handling units shall be factory insulated, double wall type in accordance with ARI 430 with sound ratings in accordance with ASHRAE 68. All sections of AHUs shall be factory built. Field built plenums are not acceptable.
- b. Fans shall be centrifugal type with airfoil blades where possible.
- c. New air handling units shall be variable volume type plenum fans with variable frequency drives.
- d. All new chilled water air handling units shall include an airside economizer where required by energy efficiency standards or economical.
- e. Coils shall be in accordance with ARI 410. Drain pans shall be stainless steel and provide positive drainage. Coil clearances shall be clearing indicated on the drawings. Design/locate the AHUs to prevent interference in this clearance to the greatest extent possible.
- f. Airflow through coils shall not exceed 450 FPM. Provide at least 16" of clearance between coils.
- g. Serpentine temperature sensors shall be used in lieu of single point, probe type.
- h. Access doors shall be provided in all sections and in between coils. Doors shall not be less than 12" in any one direction, however full section access is preferred.
- i. AHUs shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

VARIABLE AIR VOLUME (VAV) TERMINAL UNITS

- a. VAV units shall be in accordance with ARI 880. Maximum allowable air leakage rate shall be 2 percent at three-inch water gauge.
- b. Casings shall be minimum 26 gauge galvanized steel. Provide access panels where required for service and adjustment without disconnecting ducts.
- c. Boxes shall be of double wall construction. Insulation shall be a minimum of 1/2 inch, 1-1/2 lb. per foot density.
- d. Damper must maintain constant position / cfm within 5 percent of design rated cfm setting.
- e. Re-heat hot water coils shall be one-row or two-row, as required and an integral part of the box.
- f. VAV units shall be of the pressure independent type and capable of 100% airtight shut-off.
- g. Cable or "Worm-gear" style damper/valve control is not acceptable.

- h. VAVs shall have the capability to be controlled and monitored via the base EMCS system, therefor only DDC controlled boxes are allowed (no pneumatically controlled boxes). Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

UNITARY AIR CONDITIONING EQUIPMENT

PACKAGE TERMINAL UNITS

- a. Unit shall have a minimum efficiency in accordance with ASHRAE 90.1. Provide units removable from the inside of the building for servicing without removing the outside cabinet.
- b. Units shall have integral electric resistance heaters, ARI certified and UL listed.
- c. Indication and controls shall be connected to the Base EMCS system where practical. Provide alternate design (for pricing) to include monitoring/control of equipment 5 tons and smaller.

SINGLE PACKAGE UNITS

- a. Units shall have minimum efficiency in accordance with ASHRAE 90.1.
- b. Units shall be combination heating and cooling units with direct expansion cooling, gas heat and tested in accordance with ARI 210/240.
- c. When provided for rooftop installation, the units shall include factory roof curbs, flashing and transition plenum.
- d. All electrical disconnecting means must be provided outside of the unit and not integral to the unit. If the manufacturer does not offer the capability to remove this feature from the factory, a separate disconnect shall be provided.
- e. When building / space loads are over 20-tons, chilled water systems are preferred.
- f. Packaged units over 5 tons shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

SPLIT SYSTEM TYPE

- a. Minimum efficiency shall be in accordance with ASHRAE 90.1.
- b. Provide separate assemblies designed to be used together and operate in accordance with ARI 210/240.
- c. When building / space loads are over 20-tons, chilled water systems are preferred.
- d. DX split systems over 5 tons shall have the capability to be controlled and monitored via the base EMCS system. Contractor to provide any and all labor and equipment necessary to communicate with existing EMCS.

DUCTWORK

- a. All ductwork shall be installed in accordance with SMACNA HVAC Duct Construction Standards. Duct air leakage tests shall be performed.

- b. All ductwork shall be externally insulated.
- c. Kitchen grease exhaust duct shall be 16-gauge welded black steel or 18-gauge welded stainless steel.
- d. Kitchen dishwasher exhaust shall be stainless steel.
- e. Shower exhaust ductwork shall be aluminum or stainless steel.
- f. Provide sufficient number of access panels to permit easy access for cleaning of ducts.
- g. Mastic-plus-embedded-fabric systems are the preferred joining methods for all joints, seams, and connections. Do not use pressure sensitive tape unless previously approved by the Contracting Officer (CO) and Contracting Officer's Representative (COR).

CONTROL SYSTEMS

- a. All controls shall be compatible and communicate with existing Base EMCS. Existing control system is a Johnson Controls Metasys system.
- b. All new construction shall utilize DDC control systems. Pneumatic controls are not acceptable. Wireless DDC controls systems are not allowed on any projects.

TEST, ADJUSTING, AND BALANCING (TAB) FOR HVAC

- a. A Test & Balance (T&B) must be performed for all HVAC equipment/projects. This shall include but is not limited to:
 - a. Chillers
 - b. AHUs
 - c. DX equipment (greater than 5 tons)
 - d. Boilers
 - e. Fans
 - f. Pumps
 - g. Duct
 - h. VAVs/FTBs/Terminal Boxes
 - i. Diffusers/Grilles/Registers
 - j. Chilled and Heating Hot Water flow at all equipment
 - k. Heat exchangers
 - l. Energy Recovery Equipment
- b. TAB companies and their representatives for CAFB projects must be certified Associated Air Balance Council (AABC) members or certified by the National Environmental Balancing Bureau (NEBB) to perform TAB service for HVAC, water balancing and vibrations and sound testing of equipment.
- c. One or more of the applicable AABC, NEBB or SMACNA publications, supplemented by ASHRAE Handbook "HVAC Applications" Chapter 38, shall be the basis for planning, procedures, and reports.
- d. TAB contractor must supply the following for review and approval:
 - a. Systems inspection report on equipment and installation for conformance with design.
 - b. Duct air leakage test report.
 - c. Systems readiness report.
 - d. Intermediate and final TAB reports covering flow balance and adjustments, performance tests, vibration tests and sound tests.

- e. Include in final reports uncorrected installation deficiencies noted during TAB and applicable explanatory comments on test results that differ from design requirements.
- e. TAB contractor must submit the completed T&B report prior to final or partial final inspection.

COMMISSIONING OF HVAC SYSTEMS

- a. All HVAC projects shall be commissioned IAW UFC and UFGS.
- b. The level of commissioning services will differ for every project based upon complexity and needs therein. The expectation is that all HVAC systems will be commissioned, but the final requirements and scope will be coordinated with the CO/COR/Base Mechanical Engineer prior to the submission of a fee for design or construction services.
- c. The intent of providing commissioning services for HVAC are twofold: The first is to ensure CAFB HVAC projects are of sound design and are appropriate for the region and facility being served. The second is that the HVAC equipment operates as designed and IAW the sequences, schematics, and specifications.
- d. The Commissioning Firm shall be an integral part of the project from the NTP of design or construction services and shall see the project through to completion. The Commissioning Firm that provides design review of a project shall be the same that verifies the operational status and correctness of the completed construction project. Every effort shall be made to retain the Commissioning Firm for the life of the specified project, design to final completion.
- e. Provide a Commissioning Firm that is certified in commissioning by one of the following: the AABC commissioning Group (ACG); the National Environmental Balancing Bureau (NEBB); the International Certification Board/Testing, Adjusting, and Balancing Bureau (ICB/TABB), the Building Commissioning Association (BCA); the Association of Energy Engineers (AEE). The Commissioning Firm may employ a commissioning professional certified by the University of Wisconsin-Madison or the American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE) as an alternative to certification of the Commissioning Firm. The Commissioning Firm must be certified in all systems to be commissioned to the extent such certifications are available from the certifying body. Describe any lapses in certification or disciplinary action taken by the certifying body against the proposed Commissioning Firm or Lead Commissioning Specialist in detail. Any firm or commissioning professional that has been the subject of disciplinary action by the certifying body within the five years preceding contract award is not eligible to perform any duties related to commissioning. The Commissioning Firm must be a 1st tier subcontractor of the General or Prime Contractor and must be financially and corporately independent of all other subcontractors. The Commissioning Firm must employ a Lead Commissioning Specialist that coordinates all aspects of the commissioning process.
- f. Comply with the requirements of the commissioning standard under which the Commissioning Firm and Specialists qualifications are approved. When the firm and specialists are certified by BCA, AEE, ASHRAE, or the University of Wisconsin-Madison, comply with the requirements of one of the acceptable standards unless otherwise stated herein. The acceptable standards are ACG Commissioning Guideline, NEBB Commissioning Standard, SMACNA 1429, or ASHRAE 202.

LIGHTNING PROTECTION SYSTEM UFC 3-575-01System Requirements:

Materials shall consist of standard products of a manufacturer regularly engaged in production of lightning protection systems and shall be manufacturer's latest UL approved design. Lightning protection system shall conform to NFPA 70, NFPA 780, UL 96 and UL 96A. Provide stamped drawings. Provide Master Certificate. *For Air Force: The contract for a lightning protection system project, or for any project on a facility containing a lightning protection system, shall require an LPS inspection prior to acceptance. The LPS shall be inspected by a commercial, third-party inspector whose sole work is lightning protection, and shall be certified by this third-party inspector as compliant with AFI 32-1065 and NFPA 780, in that priority order. A UL certification on its own shall not be adequate for acceptance of any Air Force project. Projects calling for an LPS addition to an existing LPS project shall consider the configuration of the final LPS in the initial design. Projects of this type shall ensure the final LPS as a whole is compliant with AFI 32-1065 and NFPA 780, in that priority order. The same third-party inspector requirements shall apply.*

EXTERIOR FIRE REPORTING SYSTEM, RADIO TYPE UFC 3-600-01Radio Alarm Transmitters/Transceivers:

Provide radio alarm transmitters/transceivers completely assembled, tested at the factory, and delivered ready for installation and operation. The transmitter electronics package shall be contained within the housing as a complete assembly, removable to facilitate servicing and replacement. Transmitters shall be capable of communicating and compatible with the existing Monaco Base Wide System. Units shall have transmitting and receiving functions consistent with the existing Base Wide System.

Transmitter Identity Code:

Provide transmitters which include a unique identity code as part of each transmission. Setting the code shall be readily accomplishable in the field without the need to exchange or alter circuit boards. Obtain the code numbers from the Base Fire Department serving the facility.

Memory:

Provide transmitters with memory capability. Simultaneous or subsequent actuation of zones, including those actuated during "off air" periods, shall not result in the loss of any messages. All messages shall be stored until they are transmitted.

Automatic Test:

Provide transmitters which automatically transmit a test signal at least once in each 24 hour period or when polled. Transmitters shall also allow manual actuation of test signal by a secured (not publicly accessible) means.

ADDRESSABLE INTERIOR FIRE ALARM SYSTEMRepair Service/Replacement Parts:

Repair services and replacement parts for the system shall be furnished under the contract and be available for a period of 10 years after the date of final acceptance of the work. On-site service during the guarantee period shall be provided within 24 hours after notification. All repairs shall be completed within 48 hours after notification.

Manufacturer Qualifications:

Components shall be of current design and shall be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as otherwise or additionally specified herein and be compatible with the existing Base Wide System manufactured by Monaco.

INTERIOR FIRE ALARM SYSTEM DESIGN: Devices shall be Class A type. Class B now permitted by the UFC 3-600-01

System Operation: The system shall be a complete, supervised, coded, addressable NFPA 72 fire alarm system.

Smoke Sensors

Ionization/Photoelectric Type Smoke Sensors: Provide addressable ionization or photoelectric type smoke sensors.

Duct Smoke Sensors: Duct smoke sensors shall be addressable photoelectric type and shall be provided in ductwork in accordance with NFPA 90A and in accordance with manufacturer's recommendations.

Thermal Sensors: Thermal sensors shall be combination rate-of-rise and fixed temperature sensing.

Emergency Power Batteries: Provide sealed, maintenance-free, sealed lead acid batteries as the source for emergency power to the FACP. Batteries shall contain suspended electrolyte. The battery system shall be maintained in a fully charged condition by means of a solid state battery charger. Provide an automatic switch to transfer the load to the batteries in the event of the failure of primary power.

Fire Alarm Control Panel (FACP): Provide a complete control panel fully enclosed in a lockable steel enclosure.

History Logging: In addition to the required printer output, the control panel shall have the ability to store a minimum of 400 events in a log.

Remote Fire Alarm Control Units: Provide complete units fully enclosed in a lockable steel enclosure.

Radio Transmitter/Transceiver and Interface Panels: Provide radio transmitter with antenna that is compatible with the existing supervising station fire alarm system manufactured by Monaco. Transmitter shall have a means to transmit alarm, supervisory, polling, and trouble conditions via a single transmitter. Provide transmitters in accordance with applicable portions of NFPA 72, Federal Communications Commission (FCC) 47 CFR 90. Transmitter/receiver shall have a source of power for operation which conforms to NFPA 72. Transmitter shall be capable of initiating a test signal daily at any selected time.

FIRE SUPPRESSION SYSTEMS

Wet Pipe Fire Sprinkler System: Provide in accordance with UFC.

Design in accordance with provisions of UFC 3-600-01 and UFC 3-600-10. Base pipe sizing and distribution on Hydraulic Calculations. Provide stamped drawings and qualifications of Fire Protection Engineer doing the design.

Sprinkler heads in spaces with ceilings shall be semi-recessed type unless ceiling heights are below 8'-0" where fully recessed heads shall be used. Extended coverage heads are not acceptable without written approval.

Dry pipe fire sprinkler systems shall be used in areas where piping is subject to freezing.

All interior piping shall be Schedule 40 black steel for sizes less than 8 inches and Schedule 30 for sizes 8 inches and larger. Piping installation shall comply with ATFP Seismic requirements.

Double check valve backflow preventer assembly shall be provided for all fire sprinkler systems.

Fire Pump System: Provide in accordance with UFC and as follows:

Provide a package fire pump system and associated equipment complete and ready for operation. Equipment, inspection and testing shall be in accordance with NFPA 20, NFPA 70 and NFPA 72. Devices and equipment for fire protection service shall be UL Fire Protection Directory Listed.

The fire pump system shall be a fully assembled and tested package system including fire pump, pressure maintenance "Jockey" pump, control panel, fuel system (if required), valves and all appurtenances required for a complete fire pump system in accordance with NFPA 20.

Provide posted operating and testing instructions and signage as required by NFPA.

Control panel and alarm devices shall be compatible and communicate with the existing base fire alarm system.

MASS NOTIFICATION SYSTEM: Provide in accordance with UFC and the following:

A mass notification system shall be installed in accordance with UFC 4-021-01. The system shall incorporate visual and audible signals and shall be capable of providing live announcements. The system shall be interconnected with the fire alarm system and the public address system. The existing system is a Monaco BT-XF transmitter with public address amplifiers and combination visual/audible devices.

*** END ***